

POWER STATION ENGINEERING – Assignment Questions

Module-I

1. What are the uses of load curves in a power system?
2. Define plant utilization factor.
3. Define Plant use factor factor.
4. Define plant capacity factor.
5. A power station has to supply load on daily basis as follows :

Time (hours)	0 - 6	6 - 8	8 - 12	12 - 14	14 - 18	18 - 20	20 - 24
Load (kW)	4,500	3,500	7,500	2,000	8,000	2,500	5,000

Sketch the load – duration curve and determine the load factor and plant capacity factor if the capacity of the plant is 12 MW.

6. Prove that the load factor of a power system is improved by an increase in diversity of load.
7. A power station has maximum demand of 80 MW, a load factor of 0.7, plant capacity factor of 0.5 and plant use factor of 0.90. Find (i) the daily energy produced, (ii) the reserve capacity of the plant, and (iii) the maximum energy that could be produced daily if the plant operating schedule is fully loaded when in operation.
8. If a generating station has a maximum load for the year of 18,000 kW and a load factor of 30.5% and the maximum loads on the substations were 7500 kW, 5000 kW, 3400 kW, 4600kW and 2800 kW. Calculate the units generated for the year and diversity factor of the generating station.
9. The maximum demand on a power station is 50,000 kW. If the average load is 20,000kW, find load factor.
10. A generating station has a connected load of 60 MW and a maximum demand of 30 MW. Units generated are 92 MWh per year. Determine the load factor.
11. The yearly duration curve of a certain power plant can be considered as a straight line from 300 MW to 80 MW. Power is supplied with one generating unit of 200 MW capacity and two units of 100 MW capacity each. Determine:
 - (i) Installed capacity
 - (ii) Load factor
 - (iii) Plant capacity factor
 - (iv) Maximum demand
 - (v) Utilization factor
12. What is the significance of load curves?
13. What is spinning reserve unit?
14. What is hot reserve unit?
15. What is cold reserve unit?

16. What do you understand by the term tariff?
17. What are different types of tariff? Give the economic basis for adapting two part tariff.
18. What are different types of tariff? Give the economic basis for adapting three part tariff.
19. A consumer has a yearly consumption of 200000 units. The tariff is Rs 50 per annum per kW of maximum demand + 10 paisa / kWh. Find the total annual cost and overall cost per unit if
 - (i) consumption were 200000 units per annum with load factor of 25%
 - (ii) consumption were reduced by 25% with same load factor.

Module – III (Hydro Power Plant)

1. Define flow duration curve.
2. What is the utility of flow duration curve?
3. What is a mass curve?
4. What does the slope of the mass curve at a point indicate ? How is the reservoir capacity determined with the help of a mass curve?
5. What does the slope of the mass curve at a point indicate?
6. How is the reservoir capacity determined with the help of a mass curve?
7. What is the main purpose of the reservoir?
8. State the function of a dam. How are dams classified? Briefly describe a few important types of dams.
9. Why trash rack is used?
10. How a surge tank helps in reducing water hammer effect in a hydroelectric power plant?
11. Explain about penstock?
12. What is the use of spill ways?
13. Write about prime movers?
14. What is the use of draft tube?
15. What are the three main factors for power output of hydroelectric plant?
16. Discuss the essential factors which should be considered while selecting a site for a hydroelectric power plant?
17. What is the function of a hydraulic turbine? What are the basic differences between impulse and reaction turbines?
18. What do you understand by the term specific speed of a water turbine?
19. Write a brief note on plant auxiliaries of hydro power plant.
20. State the advantages and disadvantages of the hydroelectric power plants.
21. Differentiate pelton wheel turbine with Francis turbine.
22. Differentiate storage with pondage.

23. Give an example for a low head turbine, a medium head turbine and a high head turbine in hydroelectric power plant.
24. Explain the various factors to be considered in the selection of a hydraulic turbine.
25. A river has a discharge of 6000 cusecs (cubic meters/sec) and the head of water available for power generation is 65 meters. Assuming the overall efficiency as 80%, Find the quantum of electrical power that can be generated from a hydro electric project.
26. A hydroelectric plant produces 10 MW under a head of 20m. Determine the specific speed of the turbine, if it has to run at a speed of 148 r.p.m.
27. The following data relates to a hydroelectric power plant:
 Head = 25m, River discharge = 400 cusecs (m^3/s), Turbine efficiency = 88%,
 Frequency of generation = 50 Hz, Number of poles = 24
 The generator is directly coupled to turbine. Calculate the least number of turbines required if
 (i) Francis turbine is used with a specific speed of 300 rpm
 (ii) Kaplan turbine is used with a specific speed of 750 rpm
28. The following data is available for a hydro power plant:
 Head = 150m, Catchment area = 2000 km^2 , Average annual rainfall = 145cm,
 Turbine efficiency = 85%, Generator efficiency = 90%, Percolation and evaporation losses = 20%
 Determine following:
 (i) Power developed,
 (ii) Suggest the type of turbine to be used if the runner speed is to be kept below 240 rpm.
29. The following data relates to a hydroelectric power plant:
 Head = 400 m
 Discharge = 4.5 m^3/s
 Turbine efficiency = 82 %
 Specific speed = 60
 Determine the power developed, the type of turbine and the speed of the turbine.

30.

- (a) What is mass curve? How is the reservoir capacity determined with the help of a mass curve?
- (b) The annual runoff in terms of depth over the catchment area of 1680 km^2 of a reservoir is tabulated as given below:

Year	Runoff (cm)	Year	Runoff (cm)
2001	98	2005	96
2002	143	2006	153
2003	168	2007	110
2004	95	2008	131

Draw the mass inflow curve. What is the average yield from the catchment area?

31. The following data relate to a proposed hydro electric station:

Available head = 28m; Catchment area = 420 km² ; Rainfall = 140 cm/year ; Percentage of total rainfall utilized = 68% ; Penstock efficiency = 94% ; Turbine efficiency = 80% ; Generator efficiency = 84% and load factor = 44% . Calculate the power developed in kW .

32. What is the function of a hydraulic turbine? What are the basic differences between impulse and reaction turbines?

33. The following data is available for a hydro power plant:

Head = 150m, Catchment area = 2000 km², Average annual rainfall = 145cm, Turbine efficiency = 85%, Generator efficiency = 90%, Percolation and evaporation losses = 20%

Determine following:

(i) Power developed,

(ii) Suggest the type of turbine to be used if the runner speed is to be kept below 240 rpm.